

Statistics for Scientific Claims

Unit 1

Data, Measurement, and Uncertainty

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Data types, measurement quality, uncertainty, error, replication, and responsible scientific language.
Central habit	Before choosing a statistical test, identify what kind of data you have and how trustworthy those data are.

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1 Purpose of Unit 1

This unit establishes the foundation for all later statistical reasoning. Students often treat statistics as a collection of formulas or software commands. That is backwards. Statistics begins with the quality and structure of the data.

A scientific investigation does not produce conclusions directly. It produces measurements. Those measurements are shaped by instruments, procedures, biological variation, human choices, sample size, and the design of the experiment. The first statistical skill is therefore not calculation. It is diagnosis.

Core thesis

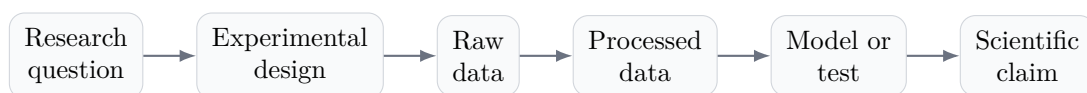
A scientific conclusion is not simply “what happened.” It is a controlled claim about what the data support, given uncertainty.

By the end of this unit, students should be able to:

1. distinguish categorical, ordinal, discrete count, continuous, and proportion data;
2. explain why data are measurements with uncertainty, not perfect facts;
3. distinguish accuracy, precision, random error, and systematic error;
4. identify instrument resolution and repeated-measure variation as sources of uncertainty;
5. distinguish technical replicates from biological replicates;
6. decide whether an outlier is likely an error, a rare value, or real biological variation;
7. write cautious scientific statements instead of overconfident claims.

2 The evidence chain

Statistics sits inside a larger chain of scientific reasoning:



Each step can fail. A strong p-value cannot rescue a vague research question, poor measurement technique, non-independent samples, or uncontrolled confounding variables.

Common failure

Students often jump from a visible pattern to a scientific conclusion. Example: “The fertilizer worked because the plants were taller.” This is incomplete. The correct questions are: How much taller? Compared with what? How variable were the measurements? Were the plants independent? Was the difference large enough to matter biologically?

3 Data are not conclusions

A table of values is not yet evidence. A table becomes evidence only after we understand what the values represent.

Consider this simple dataset:

Group	Plant 1	Plant 2	Plant 3	Plant 4	Plant 5
Control height / cm	11.8	12.6	12.1	13.0	12.5
Fertilizer height / cm	14.2	15.8	13.9	16.4	15.2

A weak conclusion would be:

The fertilizer improved growth because the fertilizer group had higher values.

A stronger statistical beginning would be:

The fertilizer group had a higher mean plant height than the control group. However, the strength of the conclusion depends on the amount of variation within each group, sample size, independence of the plants, and control of environmental factors.

The second version is better because it does not pretend that the table alone proves the biological mechanism.

4 Types of data

The type of data controls the possible analysis. Students should identify data type before choosing a graph or statistical test.

Data type	Meaning	IB science examples	Typical displays
Categorical nominal	Named categories with no natural order	blood type, phenotype, color category, alive/dead	bar chart, contingency table
Ordinal	Categories with a natural order, but unequal spacing	pain score, disease severity scale, qualitative reaction intensity	ordered bar chart, rank summary
Discrete count	Whole-number counts	colony count, number of germinated seeds, offspring phenotype counts	bar chart, dot plot, count table
Continuous measurement	Measured values on a scale	mass, height, time, concentration, absorbance, pH	dot plot, box plot, histogram, scatter plot
Proportion or percentage	Part of a whole, often derived from counts	survival rate, germination percentage, percentage cover	bar chart with counts shown, confidence interval

Practical rule

Never report only percentages for categorical results. Always report the counts behind them. A survival rate of 80% from 10 organisms is not the same strength of evidence as 80% from 1000 organisms.

4.1 Examples of correct identification

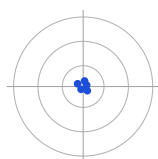
Research question	Outcome variable	Data type
Does sodium chloride concentration affect seed germination?	Number germinated out of total seeds	Count/proportion
Does temperature affect catalase activity?	Oxygen volume produced per minute	Continuous measurement
Do pea offspring fit a 3:1 ratio?	Phenotype category counts	Categorical counts
Does pH affect amylase activity?	Time for starch disappearance	Continuous measurement, often skewed
Does treatment affect bacterial survival?	Alive/dead counts	Binary categorical/proportion

5 Measurement quality: accuracy and precision

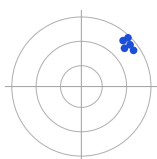
Accuracy means closeness to the true or accepted value.

Precision means closeness of repeated measurements to each other.

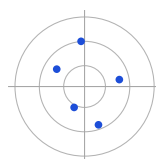
These are different. A method can be precise but inaccurate if it gives consistent values that are all shifted away from the true value.



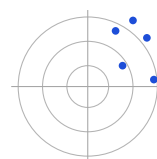
accurate and precise



inaccurate but precise



accurate on average, not precise



inaccurate and not precise

Example: titration

Suppose the accepted concentration of sodium hydroxide is $0.1000 \text{ mol dm}^{-3}$. A student repeatedly obtains 0.0921 , 0.0924 , and $0.0922 \text{ mol dm}^{-3}$. The results are precise because they are close to each other. They are inaccurate because they are far from the accepted value. A possible systematic cause is incorrect standardization or a consistently overshoot endpoint.

6 Random error and systematic error

Random error causes measurements to scatter unpredictably. It affects precision. Random error can often be reduced by repeated independent measurements and improved technique.

Systematic error shifts measurements consistently in one direction. It affects accuracy. Systematic error is more dangerous because repeated measurements may look consistent while still being wrong.

Error type	What it does	Examples
Random error	Produces scatter around a value	endpoint judgement varies, small timing differences, pipetting variability, biological variation between organisms
Systematic error	Shifts values in one direction	uncalibrated balance, contaminated reagent, consistent parallax error, colorimeter not blanked, temperature consistently above target
Gross error	A mistake or failure that invalidates a measurement	sample spilled, wrong reagent used, impossible recorded value, mislabelled tube

Do not write “human error”

“Human error” is too vague for serious scientific evaluation. Replace it with the specific mechanism: endpoint overshoot, incomplete mixing, delayed timing, sample contamination, incorrect dilution, instrument not zeroed, evaporation, temperature drift, or mislabelled sample.

7 Uncertainty

Uncertainty is not the same as error. Error is the difference from the true value, which is often unknown. Uncertainty is our estimate of the plausible range around a measurement or result.

Core distinction

A measurement without uncertainty is incomplete. The uncertainty tells the reader how much trust to place in the numerical value.

7.1 Instrument resolution

Instrument resolution is the smallest scale division or display increment.

Instrument	Typical reading	Typical uncertainty note
Digital balance, 0.01 g resolution	12.36 g	Often ± 0.01 g, unless calibration or drift is worse
Ruler, 1 mm divisions	42 mm	Often ± 0.5 mm for careful reading; more if endpoints are unclear
Volumetric pipette, 25.00 mL	25.00 mL	Use manufacturer tolerance when available
Burette, 0.10 mL divisions	Initial and final readings	Each reading has uncertainty; delivered volume depends on two readings
Colorimeter	absorbance value	Instrument precision, blanking, cuvette cleanliness, and calibration curve dominate

The instrument scale is only one source of uncertainty. In biology, biological variation can be much larger than instrument resolution.

7.2 Repeated measurements

Repeated measurements help estimate variation. But the interpretation depends on what was repeated.

Example: repeated titration volumes

A student obtains concordant titres: 24.60, 24.65, and 24.55 mL. These repeated values suggest good procedural precision. The mean titre is:

$$\bar{x} = \frac{24.60 + 24.65 + 24.55}{3} = 24.60 \text{ mL.}$$

The range is 0.10 mL. This supports precision, but not necessarily accuracy. If the standard solution was incorrectly prepared, all titres could still be systematically wrong.

7.3 Absolute and percentage uncertainty

For a measurement $x \pm \Delta x$:

$$\text{percentage uncertainty} = \frac{\Delta x}{x} \times 100\%.$$

For example, if a volume is recorded as 25.00 ± 0.05 mL, then:

$$\frac{0.05}{25.00} \times 100\% = 0.20\%.$$

The same absolute uncertainty is more serious for a small measurement than for a large measurement.

Measurement	Absolute uncertainty	Percentage uncertainty
2.00 mL	± 0.05 mL	2.5%
25.00 mL	± 0.05 mL	0.20%
100.00 mL	± 0.05 mL	0.05%

8 Raw data and processed data

Raw data are direct observations or measurements.

Processed data are values calculated from raw data.

For example, in a titration:

Raw data	initial burette reading, final burette reading, temperature, qualitative endpoint observation
Processed data	titre, mean titre, concentration, percentage uncertainty, graph slope

Students should avoid presenting only processed results. The reader needs to see enough raw data to judge whether processing was legitimate.

Common IA problem

A student reports only the final calculated concentration but hides the original burette readings. This prevents the reader from evaluating concordance, endpoint consistency, and possible transcription errors.

9 Significant figures and responsible rounding

Significant figures are not decoration. They communicate measurement precision.

9.1 Basic reporting rule

A result should not be reported with more precision than the measurement supports.

Bad:

$$\text{Mean mass} = 12.3466666667 \text{ g.}$$

Better:

$$\text{Mean mass} = 12.35 \text{ g.}$$

If uncertainty is reported, round the uncertainty sensibly and round the value to the same decimal place:

$$12.35 \pm 0.04 \text{ g.}$$

9.2 Why over-rounding is also bad

Over-rounding destroys information. If a colorimeter gives absorbance values of 0.412, 0.426, and 0.419, reporting all as 0.4 hides real variation.

The goal is not to minimize digits. The goal is to report digits justified by the method.

10 Replication

Replication is one of the most important ideas in experimental science. But not all repeats mean the same thing.

10.1 Technical replicates

Technical replicates are repeated measurements of the same biological or chemical sample. They estimate measurement precision.

Examples:

- measuring the same absorbance sample three times;
- weighing the same object repeatedly;
- running the same PCR product twice from the same tube;
- taking several readings from the same bacterial culture.

Technical replicates are useful, but they do not create independent biological evidence.

10.2 Biological replicates

Biological replicates are independent biological units or independent experimental samples.

Examples:

- different plants grown under the same condition;
- independent bacterial cultures started separately;
- different organisms, patients, leaves, colonies, or tissue samples;
- independent reaction mixtures prepared separately.

For biological inference, biological replicates matter more.

The false sample size problem

If a student measures the same bacterial culture ten times, the biological sample size is not $n = 10$. It is still $n = 1$ biological replicate with ten technical readings. Treating technical readings as independent biological samples creates false confidence.

Example: enzyme activity

A group prepares one catalase extract and measures oxygen production five times. This tests repeatability of the measurement procedure. It does not show how catalase activity varies across different biological samples.

A stronger design prepares catalase extract from five independent biological samples and measures each sample once or twice. This allows a claim about biological variation, not only instrument repeatability.

11 Outliers

An outlier is a value that appears unusual compared with the rest of the data. It must not be automatically deleted.

A useful diagnostic question is:

Is this value impossible, procedurally compromised, rare but plausible, or biologically meaningful?

Outlier type	Interpretation	Action
Impossible value	Contradicts physical or procedural reality	Exclude with explanation; check records
Procedural failure	Sample spilled, wrong reagent, contaminated cuvette	Exclude only if documented before seeing desired conclusion
Rare but plausible value	Extreme biological variation or unusual measurement	Usually keep; discuss uncertainty
Influential value	Strongly changes slope, mean, or conclusion	Analyze with and without it, justify treatment

Do not delete inconvenient data

Removing outliers because they weaken a conclusion is p-hacking. Outlier rules should be defined before analysis where possible, or justified by documented procedural evidence.

12 Worked examples

12.1 Worked example 1: titration data

A student standardizes sodium hydroxide using titration. The titres are:

Trial	Titre / mL
Rough	25.80
1	24.65
2	24.60
3	24.70

The rough trial should not usually be included in the final mean because it is used to locate the endpoint region. The concordant trials are 1, 2, and 3.

$$\bar{x} = \frac{24.65 + 24.60 + 24.70}{3} = 24.65 \text{ mL.}$$

A good comment would be:

The three final titres were concordant within 0.10 mL, suggesting good procedural precision. This does not by itself prove accuracy, because systematic errors such as incorrect standard solution concentration or endpoint overshoot may still be present.

12.2 Worked example 2: seed germination

A student tests whether salt affects seed germination. They use 20 seeds in control and 20 seeds in salt solution.

Condition	Germinated	Not germinated	Total
Control	18	2	20
Salt solution	12	8	20

The outcome is categorical count data, not continuous measurement data. The processed values are:

$$\text{Control germination} = \frac{18}{20} \times 100\% = 90\%,$$

$$\text{Salt germination} = \frac{12}{20} \times 100\% = 60\%.$$

A responsible statement:

The salt condition showed lower germination in this sample, with a difference of 30 percentage points. Because the data are counts from a limited sample, further analysis should use a method appropriate for proportions or categorical counts, and the conclusion should remain cautious.

12.3 Worked example 3: colorimeter data

A student measures absorbance for one standard solution three times:

Reading 1	Reading 2	Reading 3	Mean
0.412	0.426	0.419	0.419

These readings are technical replicates of one solution. They help estimate instrument/procedural precision for that solution. They do not replace independent standards in a calibration curve.

A stronger calibration design includes several concentrations, prepared independently where possible, each measured with careful blanking and consistent cuvette handling.

13 Language for scientific claims

Students need disciplined language. The table below shows common weak phrases and stronger replacements.

Weak phrase	Better phrase
The hypothesis was proven.	The data support the hypothesis under the tested conditions.
There was no effect.	No statistically significant effect was detected.
Human error affected the results.	Endpoint overshoot may have increased the measured titre.
The results are accurate.	The results were precise; accuracy is limited by calibration and possible systematic error.
The graph shows a relationship.	The scatter plot suggests a positive association within the tested range.
The experiment was reliable because repeats were done.	The repeated measurements were consistent, but the independence of replicates must be considered.

Model sentence

The data suggest ..., but the strength of this conclusion is limited by The main uncertainty comes from ..., and a stronger design would

14 Mini-practice tasks

Task 1: Identify data type

For each outcome, identify the data type.

- a) Number of bacterial colonies on each agar plate.
- b) Absorbance of copper sulfate solutions.
- c) Blood type of patients.
- d) Number of seeds germinated out of 30.
- e) Disease severity recorded as mild, moderate, or severe.

Task 2: Accuracy and precision

A student measures a standard mass of 10.00 g and obtains 9.42, 9.44, and 9.43 g. Describe accuracy and precision. Suggest one possible systematic cause.

Task 3: Technical or biological replicates

Classify each case.

- a) The same cuvette is measured five times in a colorimeter.
- b) Five independent plant seedlings are grown in the same fertilizer concentration.
- c) One bacterial culture is sampled three times from the same flask.
- d) Three independent bacterial cultures are prepared from separate inoculations.

Task 4: Better scientific language

Rewrite the following statements in a more defensible form.

- a) "There was no effect of temperature."
- b) "The hypothesis was correct."
- c) "Human error caused bad results."
- d) "The experiment was accurate because the repeats were close."

Task 5: Uncertainty

A volume is measured as 10.00 ± 0.05 mL. Calculate the percentage uncertainty. Then compare it with 50.00 ± 0.05 mL.

15 Answer key

Task 1

- a) Discrete count.
- b) Continuous measurement.
- c) Categorical nominal.
- d) Count/proportion.
- e) Ordinal.

Task 2

The results are precise because the repeated values are close to each other. They are inaccurate because they are far below the accepted value of 10.00 g. A possible systematic cause is an uncalibrated balance, incorrect zeroing, or a consistent container/tare problem.

Task 3

- a) Technical replicate.
- b) Biological replicates.
- c) Mostly technical or pseudo-replicates, because the same culture is the biological source.
- d) Biological replicates.

Task 4

Possible rewrites:

- a) No statistically significant effect of temperature was detected under the tested conditions.
- b) The data support the hypothesis under the tested conditions, but further evidence would be needed for a stronger claim.
- c) Endpoint overshoot, delayed timing, incomplete mixing, or another specified mechanism may have affected the results.
- d) The repeats were precise, but accuracy also depends on calibration, systematic error, and comparison with an accepted value.

Task 5

For 10.00 ± 0.05 mL:

$$\frac{0.05}{10.00} \times 100\% = 0.50\%.$$

For 50.00 ± 0.05 mL:

$$\frac{0.05}{50.00} \times 100\% = 0.10\%.$$

The same absolute uncertainty is less serious for the larger volume.

16 Unit 1 summary

What students should carry forward

Before using any statistical test, ask:

1. What kind of data do I have?
2. How were the data generated?
3. What uncertainty and error sources are present?
4. Are the replicates truly independent?
5. What claim can the data support without overclaiming?

The next unit should build from this foundation into descriptive statistics and visualization: mean, median, standard deviation, standard error, confidence intervals, dot plots, box plots, histograms, and responsible error bars.